

Characterizing a Low-Dimensional Redshift-Dependent Difference Between CMB- and Distance-Compatible Expansion Histories

Jake Enholm

October 7, 2026

Abstract

Under the endpoint constructions examined here, low-redshift distance data and Planck-calibrated references favor substantially different best-fit expansion histories: neither side fits the full DESI DR2, Pantheon+SH0ES, and Planck TT/TE/EE likelihood contract at once, and none of the tested endpoints simultaneously supplies a complete fit to all sectors. We compare three fixed endpoints over the constrained domain $z \leq 1.8$: a phenomenological low-redshift distance-sector endpoint R reaching $H_0 = 72.80 \text{ km s}^{-1} \text{ Mpc}^{-1}$, a Planck-calibrated reference C1 with $H_0 = 67.99 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and a secondary Planck-compatible reference C2 with $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (a separately implemented sensitivity check sharing the same Planck sky, not an independent confirmation). Within the predeclared reconstruction family, constrained-domain predictive validation (SN plus DESI $z \leq 1.8$ folds; Ly α scored separately as a failed continuation test) selects a compact five-coordinate order-3 map of the endpoint difference with local projected Jacobian rank 5 of 5 fitted coordinates ($\chi^2 = 1459.96$, only 8.04 above the 21-coordinate control). The fixed order-3 map improves held-out supernova prediction by $\Delta\chi^2 = -76.1$ against the null reconstruction (the undeformed Λ CDM reference, with zero fitted reconstruction coordinates and the common SN nuisance parameter profiled normally) and improves constrained DESI ($z \leq 1.8$) by $\Delta\chi^2 = -6.90$ (null reconstruction $\chi^2 = 18.08$ against $m = 3$ $\chi^2 = 11.17$); the order-3 reconstruction fails its separate $z = 2.33$ Ly α continuation test strongly, and the matched constrained-domain null bootstrap (250 replicates; same null/candidate family, data split, bounds and selection rule) gives $P(\text{select} \geq m=3) = 0.216$ with Ly α continuation-failure fraction 0.788 recorded separately. Zero ruler-clock split is allowed ($\Delta\chi^2(\epsilon=0) = 0.124$). The reference-sensitivity check shows that the difference between R and C1 and the difference between R and C2 share one dominant direction (cosine 0.9997, principal angle 1.5° , common-mode fraction ≥ 0.998 per interval, largest amplitude near $z \approx 0.65$). Across the constrained redshift bins, the RMS difference between C1 and C2 is approximately 1% to 8% of the corresponding RMS difference between R and either Planck reference. In plain terms, the distance-sector endpoint has a systematically higher $H(z)$ than the Planck-calibrated references, and the size of that difference changes with redshift rather than behaving like a single constant H_0 offset. A five-coordinate order-3 reconstruction captures this redshift-dependent pattern over $z \leq 1.8$, with its largest endpoint separation near $z \approx 0.65$; the separate Ly α continuation test at $z = 2.33$ fails.

1 Introduction

The Hubble tension is an observational disagreement between the local supernova distance ladder, $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [1], and the Planck 2018 TT/TE/EE spectrum, $67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ in flat Λ CDM [2], reviewed in Verde et al. [3] and Knox and Millea [4]. The mismatch extends beyond a single number: DESI DR2 BAO [5] sharpens the late-time distance-redshift relation while the Pantheon+SH0ES compilation [6, 7] maps the supernova expansion

history, and neither a distance-calibrated nor a CMB-calibrated history fits all sectors at once. Observed values, uncertainties, and covariance matrices are held fixed across model comparisons; only theory predictions change between references.

This paper asks whether the disagreement can be represented as a compact redshift-dependent difference between separately calibrated background histories. Rather than fitting another cosmological model with additional dynamical freedom, we test whether the observable gap between already-calibrated endpoints has low-dimensional, predictable structure.

The experimental design uses three fixed endpoints. The distance-compatible history R reaches $H_0 = 72.80 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and fits DESI DR2 BAO and Pantheon+SH0ES supernovae. The CMB-compatible history C1 is a Planck TT/TE/EE-compatible reference with $H_0 = 67.99 \text{ km s}^{-1} \text{ Mpc}^{-1}$ produced with the stated CAMB/likelihood configuration. The secondary Planck-compatible reference C2 is a separately implemented baseline with $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, built through its own likelihood and spectral-parity path for reference-sensitivity checking. C1 and C2 are separately implemented in pipeline/model space, not statistically independent in data space: they share the Planck sky but differ in construction pipeline, parameter choices, and calibration route.

The reconstruction differs from a conventional cosmological-model fit in four ways. First, the endpoints are constructed separately, before any comparison is made. Second, reconstruction dimensionality is selected with held-out prediction, not training χ^2 . Third, higher flexibility is allowed to fail: overfitting orders and boundary-limited families are retained as fixed diagnostic controls. Fourth, the reference-sensitivity test changes the CMB reference history without changing the hypothesis: the same compact map must describe both Δ_1 and Δ_2 .

The headline result, over the constrained domain $z \leq 1.8$, is

$$\Delta_1(z) \simeq \Delta_2(z), \quad \|\Delta_C\| \ll \|\Delta_{1,2}\|,$$

where the fields $\Delta_1(z) \equiv \ln H_R(z) - \ln H_{C_1}(z)$, $\Delta_2(z) \equiv \ln H_R(z) - \ln H_{C_2}(z)$, and $\Delta_C(z) \equiv \ln H_{C_1}(z) - \ln H_{C_2}(z)$ are defined in Section 2.4 and summarized in the nomenclature table (Table 1). The difference between R and C1 and the difference between R and C2 share one dominant direction (cosine 0.9997, principal angle 1.5°), peak near $z \approx 0.65$, and transfer across references with fractional residual degradation 0.068, whereas the difference between C1 and C2 is much smaller throughout (approximately 1.3% to 8.4% of the corresponding R-versus-CMB difference across the constrained bins).

In plain terms, the distance-sector endpoint has a systematically higher expansion rate than the Planck-calibrated references, and the size of the mismatch changes with redshift instead of behaving like a constant H_0 shift. A compact five-coordinate reconstruction captures that pattern over $z \leq 1.8$, with its largest amplitude near $z \approx 0.65$.

The order-3 continuation does not reproduce the Ly α anchor at $z = 2.33$.

This paper identifies and characterizes that empirical deformation. The principal result of this work is therefore not a specific dynamical model, but an empirically defined expansion-history target that candidate explanations of the Planck-reference difference can be tested against. Section 2 defines the data and the R, C1, and C2 endpoints (with the nomenclature table immediately below); Section 3 the empirical reconstruction method; Section 4 the predictive model-selection tests; Section 5 the reference-sensitivity test; Section 6 the results; and Sections 7–9 the interpretation, limitations, and conclusions.

2 Data and endpoint definitions

The three expansion histories compared in this paper were obtained separately, before any comparison was made. Each is fixed: no endpoint is refit, retuned, or extended anywhere in this

analysis. All headline values resolve through generated macros to the fixed artifacts cited below. Symbols are defined once in the nomenclature table (Table 1); throughout the main text, R is the distance-compatible endpoint, C1 and C2 are the two Planck-compatible endpoints, and the order-3 representation is the selected compact reconstruction basis.

Table 1: Nomenclature: symbols used in the main analysis.

Symbol	Meaning	How obtained	Used for
R	distance-sector endpoint	fixed distance-sector endpoint	reference deformation
C_1	Planck-calibrated reference	Planck TT/TE/EE calibrated CAMB reference	CMB reference
C_2	secondary Planck-calibrated reference	separately implemented Planck/CAMB reference	sensitivity reference
Δ_1	$\ln H_R(z) - \ln H_{C_1}(z)$	derived	comparison
Δ_2	$\ln H_R(z) - \ln H_{C_2}(z)$	derived	comparison
Δ_C	$\ln H_{C_1}(z) - \ln H_{C_2}(z)$	derived	reference spread
K_H	overall amplitude of the reconstruction map	fitted (reconstruction coordinate)	compact representation
c_j	Legendre shape coefficients ($j = 0, 1, 2$)	fitted (reconstruction coordinates)	compact representation
$m = 3$	order-3 reconstruction	CV-selected	compact representation
ϵ	reciprocity parameter	profiled	diagnostic only

2.1 Distance-compatible endpoint R

R is the fixed phenomenological distance-sector endpoint used in this comparison. It reaches $H_0 = 72.80 \text{ km s}^{-1} \text{ Mpc}^{-1}$, carries $r_d = 135.39 \text{ Mpc}$, and provides the low-redshift distance-sector reference history. The compressed distance prior [8] used during endpoint discovery was validated only for conventional late-time models; it did not validate the $r_d = 135.39 \text{ Mpc}$ early-universe physics.

Concretely, R is the smooth low-redshift reference history (pinned in `paper/evidence/endpoints/endpoint_manifest.json`). It reaches $H_0 = 72.80 \text{ km s}^{-1} \text{ Mpc}^{-1}$ with $r_d = 135.39 \text{ Mpc}$ and fits the DESI DR2 and Pantheon+SH0ES distance sector. Nothing in Sections 4 or 5 depends on the history of how R was found.

2.2 CMB-compatible endpoint C1

C1 is a Planck TT/TE/EE-compatible reference background with $H_0 = 67.99 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $r_{\text{drag}} = 147.15 \text{ Mpc}$, produced with the stated CAMB/likelihood configuration. No distance-sector parameter, table, or likelihood entered its construction.

2.3 Secondary Planck-compatible reference C2

C2 is a secondary Planck-compatible ΛCDM reference with $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, used only as a reference-sensitivity check, evaluated through its own Planck likelihood. It shares the Planck sky with C1 and uses nearly the same ΛCDM background family with its own parameter choices and calibration route, so it cannot provide independent cosmological confirmation. C1 and C2 are

separately implemented in pipeline/model space, not statistically independent in data space; both use Planck information. Table 3 compares the two constructions row by row.

2.4 Common observable representation

All three endpoints are compared on one common uniform- u grid of 2000 nodes spanning $0 \leq z \leq 2.33$ (`paper/evidence/endpoints/endpoint_redshift.json`). The difference fields

$$\Delta_1(z) \equiv \ln H_R(z) - \ln H_{C_1}(z), \quad \Delta_2(z) \equiv \ln H_R(z) - \ln H_{C_2}(z), \quad \Delta_C(z) \equiv \ln H_{C_1}(z) - \ln H_{C_2}(z)$$

are evaluated on that grid and fixed in `paper/evidence/reference_sensitivity/difference_fields.npz`. The constrained domain is $z \leq 1.8$, covering all non- $\text{Ly}\alpha$ DESI blocks; the $\text{Ly}\alpha$ anchor at $z = 2.33$ tests the continuation law, not the low-redshift deformation (`paper/evidence/reference_sensitivity/domain_split.json`).

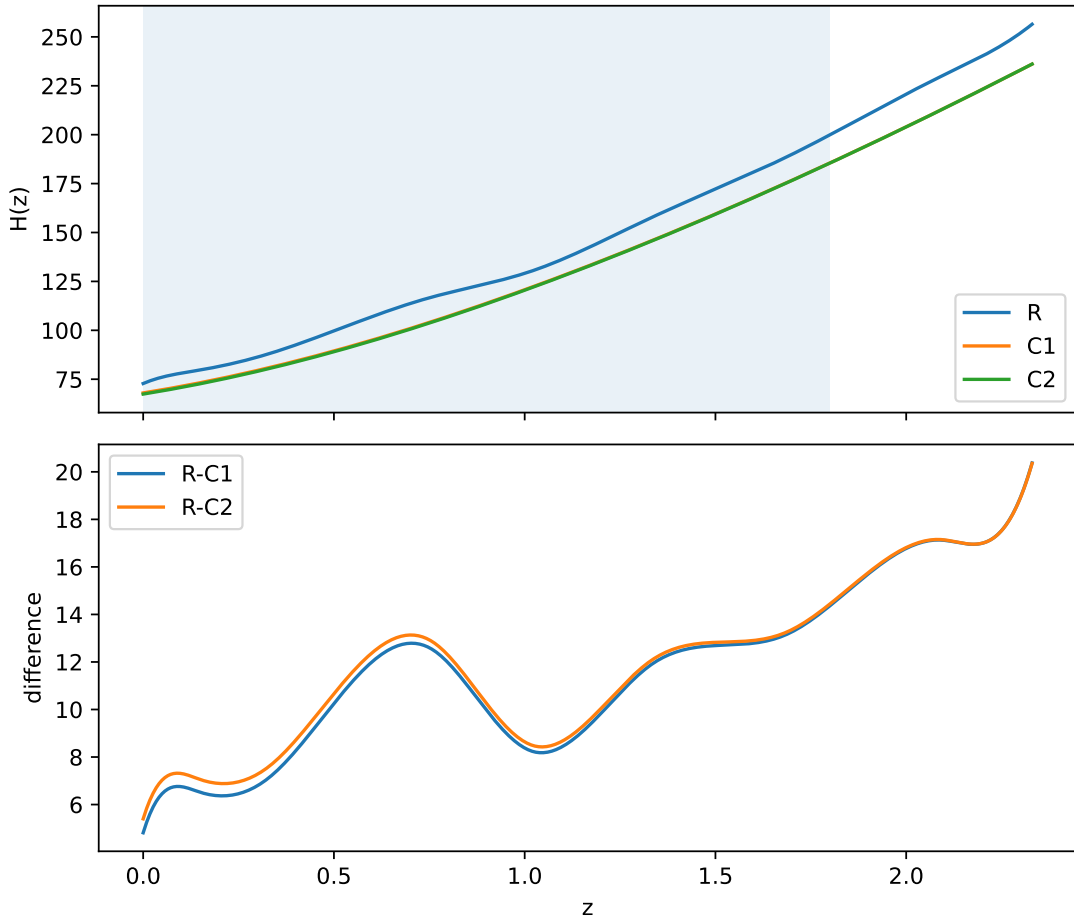


Figure 1: Fixed endpoint overview (no fits in this figure). Top: the three fixed histories H_R, H_{C_1}, H_{C_2} ; C1 and C2 nearly overlap in the upper panel and distinct line styles are used to keep both visible. Bottom: their logarithmic differences $\Delta_1, \Delta_2, \Delta_C$; across the constrained bins the RMS difference between C1 and C2 is approximately 1% to 8% of the corresponding RMS difference between R and either Planck reference. Shading marks the constrained domain $z \leq 1.8$; the dotted line marks the $\text{Ly}\alpha$ continuation test at $z = 2.33$.

2.5 Endpoint acoustic calibration

Each endpoint carries its own acoustic ruler, traced to fixed artifacts in the endpoint calibration contract (`paper/evidence/endpoints/calibration_contract.json`). Boltzmann rulers are computed with CAMB [9]:

Table 2: Endpoint acoustic calibration: every endpoint’s ruler. The compressed distance prior [8] did not validate nonstandard early-universe physics for R.

Endpoint	r_d [Mpc]	r_s [Mpc]	Fixed or recomputed?	Source
R	135.39	— (not used)	fixed (pinned control fit)	control fit
C1	147.15	144.50	recomputed (CAMB stage-0)	stage-0 background
C2	147.11	144.45	recomputed (CAMB baseline)	baseline acoustic

R uses an early-ruler value different from C1/C2; this ruler difference is part of the measured endpoint separation, not a shared calibration. The R ruler is roughly 0.92 of either CMB ruler, an $\sim 8\%$ smaller ruler. This is a property of the phenomenological distance endpoint, not a demonstrated physical sound horizon: R’s $r_d = 135.39$ Mpc is a phenomenological endpoint value and is not validated as early-universe physics. The compressed Planck distance prior [8] was validated only for conventional Λ CDM, w CDM, and CPL models and cannot by itself validate such nonstandard pre-recombination physics; no BBN viability is claimed (no BBN fit is performed). The BAO forward model is endpoint-specific:

$$D_M/r_d, \quad D_H/r_d, \quad D_V/r_d, \quad (1)$$

evaluated with each endpoint’s own ruler from Table 2.

Table 3: Endpoint construction comparison: C1 versus C2.

	C1	C2
Underlying CMB observations	Planck TT/TE/EE	Planck TT/TE/EE
Boltzmann code	CAMB 2.0.4	CAMB (baseline pipeline)
Likelihood implementation	Planck TT/TE/EE likelihood (CAMB-based)	stock baseline Planck likelihood
Cosmological parameterization	flat Λ CDM stage-0	flat Λ CDM
Primordial spectrum parameterization	power-law tilt	power-law tilt, $n_s = 0.965$
Optimizer/sampler	background tuning (stage-0)	fixed-parameter baseline with spectral parity
Background extraction	fixed CAMB table	CAMB-sampled $H(z)$ table
H_0 [km s $^{-1}$ Mpc $^{-1}$]	67.99	67.4
Ω_m	0.3075	0.3134
r_d [Mpc]	147.15	147.11
r_s [Mpc]	144.50	144.45

C1 and C2 are separately implemented in pipeline/model space, not statistically independent in data space; both use Planck information.

Observed values, uncertainties, and covariance matrices are held fixed across model comparisons. Only theory predictions change. What follows groups the likelihoods into the low-redshift distance sector, the high-redshift CMB sector, and the controls that keep the comparison honest.

2.6 Low-redshift distances

DESI DR2 BAO [5] enters through the collaboration distance ratios D_M/r_d , D_H/r_d , D_V/r_d with the published covariances, including the $z_{\text{eff}} = 2.33$ Ly α anchor [10]; DESI values are fixed and only the theory prediction changes between references.

The supernova block is the full Pantheon+SH0ES compilation [1, 6, 7] as one 1657-row covariance likelihood with one common profiled absolute magnitude M (row-selection contract, `paper/evidence/likelihood/sn_selection_contract.json`: 1701 raw rows selected by the exact mask ($z_{\text{HD}} > 0.01$) | IS_CALIBRATOR to 1657 rows, of which 77 calibrators and 1580 Hubble-flow rows). With $\mu_{\text{no } M}$ the endpoint distance modulus at $M = 0$,

$$r_0 = m_B^{\text{corr}} - \mu_{\text{no } M}, \quad (2)$$

$$\hat{M} = \frac{\mathbf{1}^T C^{-1} r_0}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad (3)$$

$$\chi_{\text{joint}}^2 = (r_0 - \hat{M}\mathbf{1})^T C^{-1} (r_0 - \hat{M}\mathbf{1}), \quad (4)$$

where C is the full 1657×1657 statistical-plus-systematics covariance. The primary joint likelihood profiles one common M (Eq. (3)). Ladder and non-ladder subsets are separately profiled for diagnostics,

$$\hat{M}_{\text{ladder}}, \quad \hat{M}_{\text{nonladder}}, \quad (5)$$

each using its marginal covariance with an independent M , so $\chi_{\text{ladder}}^2 + \chi_{\text{non-ladder}}^2 \neq \chi_{\text{joint}}^2$. These subset values are audit diagnostics (diagnostic-only) and are not additive contributions to the primary joint likelihood.

Calibrator rows carry the published Cepheid host distance $\mu_{\text{cal}} = \text{CEPH_DIST}$, on the SH0ES Cepheid period–luminosity calibration [1, 11]. Calibrator rows use the published Cepheid host distances. No additional theory-side correction is applied to calibrator distance moduli in the endpoint construction. This is part of the fixed likelihood definition used here and is not independently tested by the reconstruction analysis. The luminosity-distance construction $D_L = (1 + z)D_M$ is Etherington’s reciprocity result [12] and assumes massless photons, null geodesics, and photon-number conservation. Hubble-flow rows use z_{HD} for the transverse integral and the $z_{\text{HD}}/z_{\text{HEL}}$ prefactor $D_L = (1 + z_{\text{HEL}})D_M(z_{\text{HD}})$.

2.7 High-redshift CMB

Two CMB uses appear in this paper and must not be confused. The distance endpoint R carries the phenomenological value $r_d = 135.39$ Mpc, which is not validated as early-universe physics: the compressed Planck distance prior [8] was validated only for conventional Λ CDM, w CDM, and CPL models and cannot by itself establish the viability of nonstandard pre-recombination physics. The C1 and C2 reference constructions are calibrated instead to the full Planck TT/TE/EE spectra with foreground-marginalized likelihoods, evaluated through a Boltzmann solver at each sampled cosmology, and carry full Planck-compatible CAMB rulers near 147.1 Mpc. Sound-speed and drag-epoch conventions follow the standard baryon-loading treatment [13, 14].

2.8 Controls

Three controls keep the endpoint comparison honest. First, Λ CDM parity: the Boltzmann pipeline reproduces reference Λ CDM spectra before either endpoint is evaluated, so pipeline error cannot masquerade as endpoint separation. Second, fixed endpoint artifacts are hash-bound: every number

in Sections 6–5 is generated from versioned machine-readable evidence with source-hash binding (`paper/evidence_manifest.yaml`); generation fails on hash mismatch. Headline numerical results are generated from or hash-bound to versioned publication evidence.

3 Empirical reconstruction method

This paper fits no cosmological model. Its only fitted objects are the reconstruction coordinates mapping the fixed endpoint histories onto each other over the constrained domain $z \leq 1.8$.

3.1 Reconstruction coordinates and difference fields

The fitted reconstruction coordinates at order m are the overall amplitude K_H , the reciprocity parameter ϵ , and the Legendre shape coefficients $c_0, \dots, c_{m-1}$ (at the selected $m = 3$: five fitted coordinates). The compared fields are $\Delta_1(z) \equiv \ln H_R(z) - \ln H_{C_1}(z)$, $\Delta_2(z) \equiv \ln H_R(z) - \ln H_{C_2}(z)$, and $\Delta_C(z) \equiv \ln H_{C_1}(z) - \ln H_{C_2}(z)$ (Section 2.4).

3.2 Likelihood residual vector and SN nuisance profile

The residual vector has 1670 rows: DESI BAO rows plus 1657 Pantheon+SH0ES rows (row-selection contract, `paper/evidence/likelihood/sn_selection_contract.json`). DESI rows are whitened with the Cholesky factor of the full published covariance; SN rows are whitened with the full 1657×1657 covariance and projected to remove the whitened-ones direction, which analytically profiles the single common absolute magnitude M (Eqs. (2)–(4)). No H_0 target, penalty, or reweighting enters anywhere.

3.3 Fold definitions and constrained selection rule

Order selection uses held-out behavior on identical folds: five supernova conditional folds and six DESI leave-block-out folds at $z \leq 1.8$. The Ly α block at $z = 2.33$ [10] is scored separately as an out-of-domain continuation test and never participates in selection. The predeclared rule selects the smallest m within 2.0 of the best combined held-out score (constrained re-score `constrained_cv.json` in paper evidence).

3.4 Projected Jacobian rank

Let $r \in \mathbb{R}^{1670}$ be the whitened, nuisance-projected residual vector and $p = (\log K_H, \epsilon, c_0, c_1, c_2)$ the fitted coordinates. The projected whitened Jacobian is

$$J_{ij} = \frac{\partial r_i}{\partial p_j} s_j, \quad (6)$$

with finite-difference steps $h_j = 10^{-6} \max(1, |p_j|)$ and column scales $s = (0.4055, 0.20, 1, 1, 1)$. Its singular values are [602.38, 157.31, 102.07, 32.03, 22.31] (condition number ≈ 27), giving local projected Jacobian rank 5/5 fitted coordinates (rank-operator contract `rank_operator_contract.json` in paper evidence). These are fitted coordinates, not measured observables.

3.5 Bootstrap and information criteria

A parametric bootstrap of the entire selection pipeline under the null reconstruction (the undeformed Λ CDM reference, with zero fitted reconstruction coordinates and the common SN nuisance parameter profiled normally; null contract `null_model_contract.json` in paper evidence) calibrates order-selection frequency; it tests the complexity of the selected representation, not the existence of the numerical difference between fixed endpoint histories. Information criteria are reported only as descriptive fit-versus-complexity summaries and are not interpreted as detection significance.

Figures and tables are generated exclusively from versioned publication evidence, all SHA-pinned in the manifest `paper/evidence_manifest.yaml`; generation fails if a source hash mismatches. All reported values are best fits; no credible or confidence intervals are claimed, and every comparison is descriptive with no Wilks-theorem calibration.

4 Predictive model-selection tests

This section reconstructs the difference between the distance-compatible history R and the CMB-compatible history C1 with a compact coordinate representation. All reconstruction results resolve to versioned publication evidence in `paper/evidence/reconstruction/` through registry macros; headline numerical results are generated from or hash-bound to versioned publication evidence.

4.1 Model-selection procedure

Within the predeclared reconstruction family (shifted-Legendre shape basis with split ruler/clock calibration) and fixed likelihood contract, orders $m = 2\text{--}6$ were fitted from multiple optimizer starts with a boundary audit, a rank audit, and epsilon profiling. Constrained-domain order selection uses held-out behavior on identical folds over $z \leq 1.8$: five supernova conditional folds and six DESI leave-block-out folds at $z \leq 1.8$. The Ly α block at $z = 2.33$ is scored separately as an out-of-domain continuation test and never participates in selection (constrained re-score `constrained_cv.json` in paper evidence; no refitting — a re-score of fixed folds only).

4.2 Main result

Constrained predictive validation selects shifted-Legendre order 3 ($m = 3$) with five fitted coordinates: total $\chi^2 = 1459.96$, only 8.04 above the 21-coordinate control despite 16 fewer fitted coordinates, with local projected Jacobian rank 5 of 5 fitted coordinates after covariance whitening and nuisance projection (Section 3). Scores are tabulated with both columns:

Table 4: Constrained-domain held-out scores by order (selection column) with the Ly α continuation score alongside. The star marks the selected order under the predeclared smallest- m -within-2.0 rule. Fixed folds, re-scored; no refitting.

m	SN held-out	DESI $z \leq 1.8$ held-out	combined (selection)	DESI Ly α (continuation)
2	8201.1	12.798	8213.920	33.6
3 *	8153.1	11.174	8164.276	59.7
4	8151.9	12.252	8164.149	262.5
5	8166.8	20.010	8186.770	5448.6
6	8160.9	23.091	8184.005	19969.2

The raw constrained minimum occurs at $m = 4$, only 0.13 below the $m = 3$ score. Under the predeclared smallest-order-within-2 rule, $m = 3$ is therefore selected. Selection therefore does not depend on the Ly α continuation failure. Information criteria are reported descriptively, side-by-side: $k = 5$, $\chi^2 = 1459.96$, $\Delta\chi^2_{\text{control}} = 8.04$, $\text{AIC} \approx 1469.96$ against ≈ 1493.92 for the $k = 21$ control ($\Delta\text{AIC} = -23.96$). In words: The order-3 reconstruction increases χ^2 by 8.04 relative to the 21-coordinate control while using 16 fewer fitted coordinates; AIC therefore favors the compact representation primarily through its parsimony penalty. This is not interpreted as a detection or a calibrated probability. The measurable shape is concentrated at low to intermediate redshift (DESI improvement peak $0.4 < z < 0.8$; redshift audit in evidence).

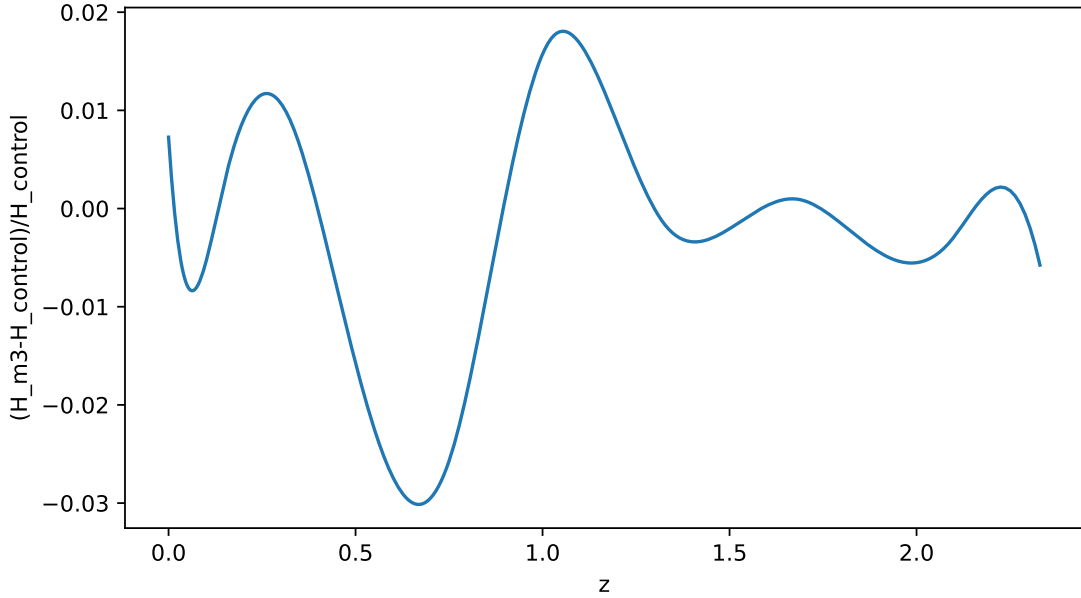


Figure 2: Five-coordinate order-3 reconstruction compared with the flexible 21-coordinate control. The lower panel shows $(H_{m=3} - H_{\text{control}})/H_{\text{control}}$.

4.3 Predictive validation and the failed continuation

Orders above the selected $m = 3$ model do not provide enough held-out improvement to overcome the predeclared parsimony rule, and still higher orders degrade predictive performance (Table 4). The null reconstruction is the undeformed ΛCDM reference with zero fitted reconstruction coordinates (K_H fixed to no deformation, ϵ not applicable) and a single common M profiled per fold-train (null contract, `paper/evidence/likelihood/null_model_contract.json`). Against this null on identical folds, the fixed $m = 3$ map improves held-out supernova prediction by $\Delta\chi^2 = -76.1$ and improves constrained DESI ($z \leq 1.8$) by $\Delta\chi^2 = -6.90$ (null reconstruction $\chi^2 = 18.08$ against $m = 3$ $\chi^2 = 11.17$); the order-3 reconstruction fails its $z = 2.33$ Ly α continuation test strongly. The Ly α block is then scored as an out-of-domain continuation test, where every order shows a large continuation error (Ly α column of Table 4). The order-3 continuation fails, and therefore the reconstruction’s constrained domain ends at $z \leq 1.8$; this bounds the claim rather than rescuing the model.

4.4 Statistical interpretation of model selection

A parametric bootstrap of the entire constrained-domain selection pipeline under the null reconstruction (250 replicates; same null/candidate family, data split, bounds and selection rule; constrained bootstrap `bootstrap.json` in paper evidence) gives $P(\text{select} \geq m=3 \mid \text{null}) = 0.216$, reported exactly as obtained with no threshold tuning. The $\text{Ly}\alpha$ continuation-failure statistic is recorded separately and never chooses m . This bootstrap tests the complexity of the selected representation, not the existence of the numerical difference between fixed endpoint histories. Model-order selection alone is not evidence of a discovery. The reference-sensitivity analysis is a separate robustness check, not a significance test. The result is presented as a compact reconstruction with demonstrated held-out behavior.

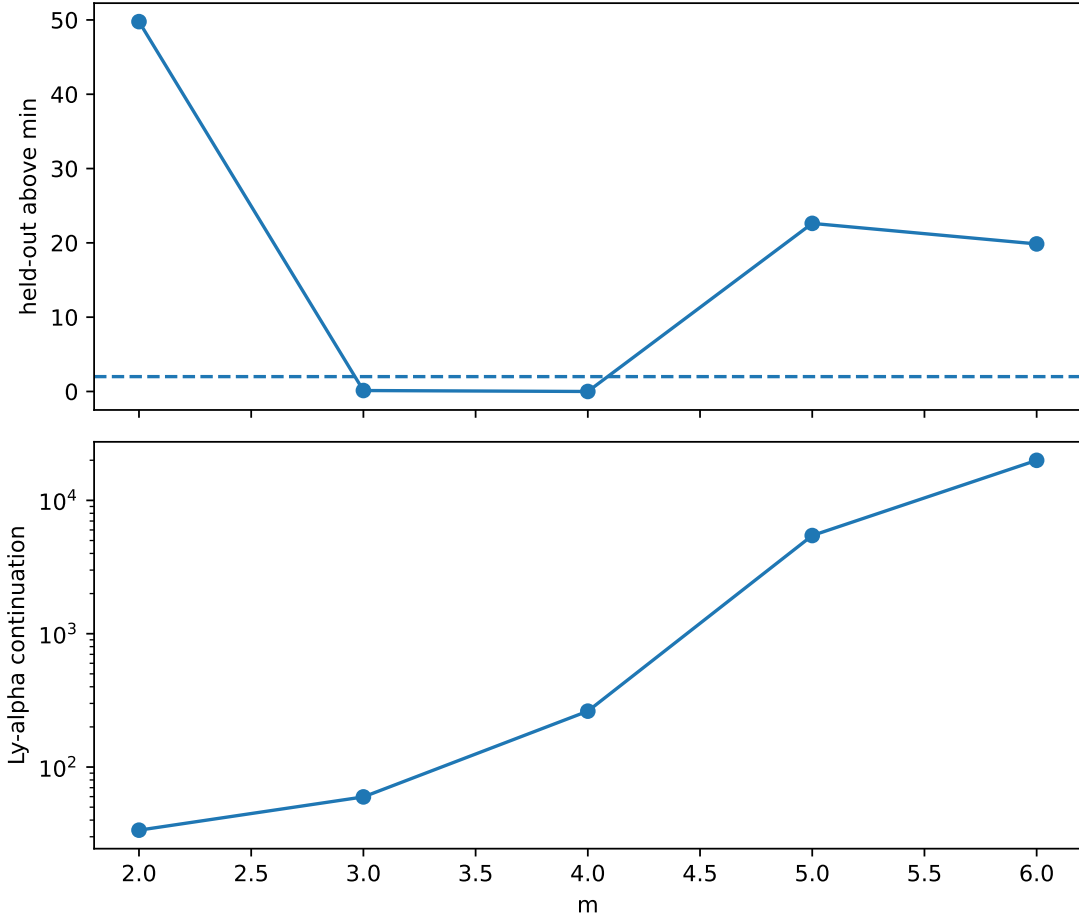


Figure 3: Constrained predictive model selection: held-out $\Delta\chi^2$ versus the constrained minimum (top; SN plus DESI $z \leq 1.8$), with the $\text{Ly}\alpha$ continuation failure on a logarithmic scale (bottom; not used in selection). The $m = 3/m = 4$ near-tie is visible; the dashed line is the within-2 selection threshold above the minimum, and the predeclared smallest- m -within-2.0 rule selects $m = 3$.

4.5 Reciprocity test

The ruler/clock split is consistent with zero: $\Delta\chi^2(\epsilon=0) = 0.124$, with 1-sigma profile approximately $[-0.012, +0.005]$. No nonreciprocal calibration is detected; ϵ is not detected.

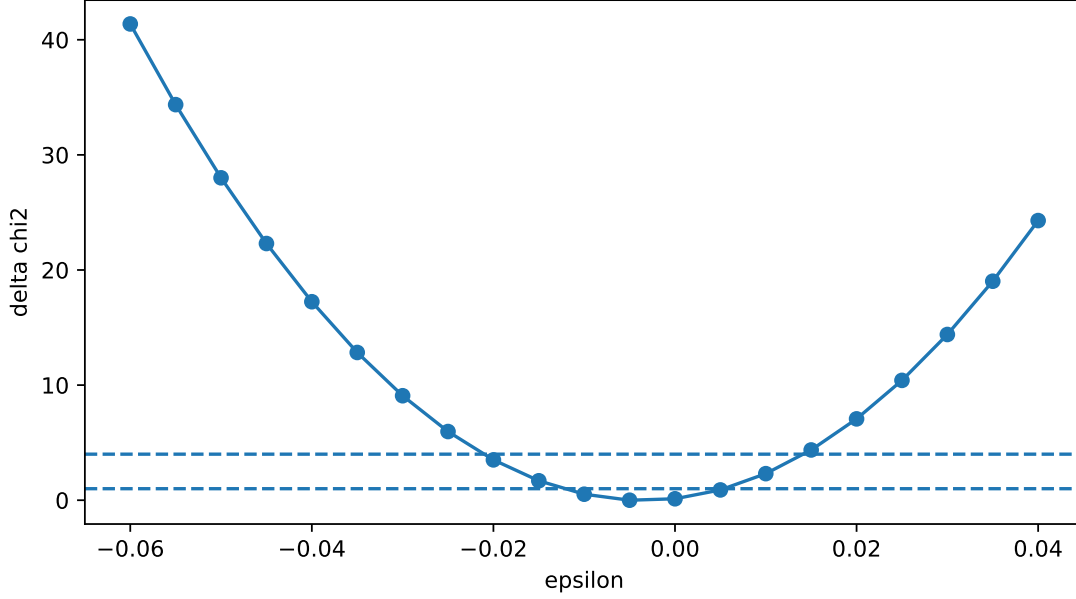


Figure 4: Reciprocity profile $\Delta\chi^2(\epsilon)$ in the selected $m = 3$ map: optimum near -0.005 and zero at $\Delta\chi^2 = 0.124$, with horizontal $\Delta\chi^2 = 1$ and 4 levels.

4.6 Why increased flexibility is rejected

The analysis actively rejected better-looking training solutions. The roughness family reaches $\chi^2 = 1453.34$ but its leading shape coefficient sits exactly at the imposed bound ($c_1 = +0.5$) with the local descent gradient pointing outward, so its optimum depends on the box: it is boundary-limited and retained as a supporting basis only. The Legendre $m = 6$ map reaches good training $\chi^2 = 1455.19$ with an interior optimum but fails the held-out Lyman-alpha continuation test (Table 4); it is therefore not promoted. Both are retained as fixed diagnostic alternatives in evidence.

5 Reference sensitivity

The reconstruction above is measured against one particular CMB-compatible history C1. This section repeats the comparison against a second, separately implemented CMB-compatible history C2 and shows the deformation is shared. All headline metrics resolve from versioned publication evidence JSON through registry macros, never handwritten.

5.1 Endpoint invariance

Define the logarithmic difference fields on the common grid (Section 2.4):

$$\Delta_1(z) = \ln H_R(z) - \ln H_{C1}(z), \quad (7)$$

$$\Delta_2(z) = \ln H_R(z) - \ln H_{C2}(z), \quad (8)$$

$$\Delta_C(z) = \ln H_{C1}(z) - \ln H_{C2}(z), \quad (9)$$

with the common and disagreement combinations

$$\Delta_{\text{common}}(z) = \frac{\Delta_1(z) + \Delta_2(z)}{2}, \quad (10)$$

$$\Delta_{\text{disagreement}}(z) = \frac{\Delta_1(z) - \Delta_2(z)}{2}. \quad (11)$$

The quoted common-mode fraction is

$$f_{\text{common}} = 1 - \left[\frac{\text{RMS}(\Delta_{\text{disagreement}})}{\text{RMS}(\Delta_{\text{common}})} \right]^2, \quad (12)$$

evaluated per redshift interval on the constrained domain. Changing the CMB reference history from C1 to C2 does not change the hypothesis: the same compact map must describe both separations.

5.2 Common-mode decomposition

On the constrained domain ($z \leq 1.8$), the difference between R and C1 and the difference between R and C2 lie substantially on top of one another. The common-mode fraction exceeds 0.998 in every redshift interval, and the two-vector singular decomposition is dominated by a single shared direction (dominance ratio 76.3). The two fields peak at the same location, near $z \approx 0.65$, while across the constrained bins the RMS difference between C1 and C2 is approximately 1.3% to 8.4% of the corresponding RMS difference between R and either Planck reference.

5.3 Peak location and data support

The fitted endpoint difference reaches its maximum near $z \simeq 0.65$. The peak does not coincide with the maximum Pantheon+ number density, supernova inverse-variance weight, or a simple DESI BAO information-density maximum. The order-3 reconstruction captures the broad low/intermediate-redshift structure but reaches its own maximum near $z \simeq 0.39$.

5.4 Cross-reference transfer

The fixed order-3 basis transfers across references: shape amplitudes refit independently agree, and parameters learned on one CMB reference predict the other separation with residual degradation fraction 0.068 ($\leq 6.8\%$ cross-reference). One dominant shared direction carries the deformation; the residual endpoint-specific component is small.

5.5 Statistical status of the reference-sensitivity test

The order-selection bootstrap tests model complexity; it does not test reference-construction similarity. The reference-sensitivity metrics quantify robustness of fixed best-fit histories: how much the measured R-versus-CMB separation moves when the CMB reference construction changes. C1 and C2 are not statistically independent (both use Planck information; Section 2.3), and the two difference fields share the same R endpoint ($\Delta_1 = R - C_1$, $\Delta_2 = R - C_2$), so their similarity is mathematically correlated. Endpoint posterior uncertainty was not propagated. Therefore no frequentist p-value is assigned to the cosine or common-mode statistic. The reference-sensitivity test shows a shared low-dimensional deformation across the tested constructions.

5.6 Constrained domain and Ly α continuation

All reference-sensitivity metrics above are quoted on the constrained domain $z \leq 1.8$, covering every non-Ly α DESI block. The Ly α anchor at $z = 2.33$ is a continuation test of the reconstruction law, not a measurement of the low-redshift deformation: flexible extrapolators generically fail there, under the null as on the data (Ly α continuation-failure fraction 0.788 in the matched constrained-domain bootstrap, recorded separately and never used in selection).

5.7 Comparison with the fit-improvement attribution

The deformation RMS (peak bin $0.4 < z < 0.8$) and the fit-improvement attribution (peak bin $0 < z < 0.4$) both concentrate below $z \sim 0.8$, though their peak bins differ; high-redshift behavior is unconstrained in both. The reference-sensitivity check therefore supports, rather than replaces, the compact reconstruction: the same five-coordinate map describes a deformation that is reference-insensitive within the two tested Planck-compatible constructions.

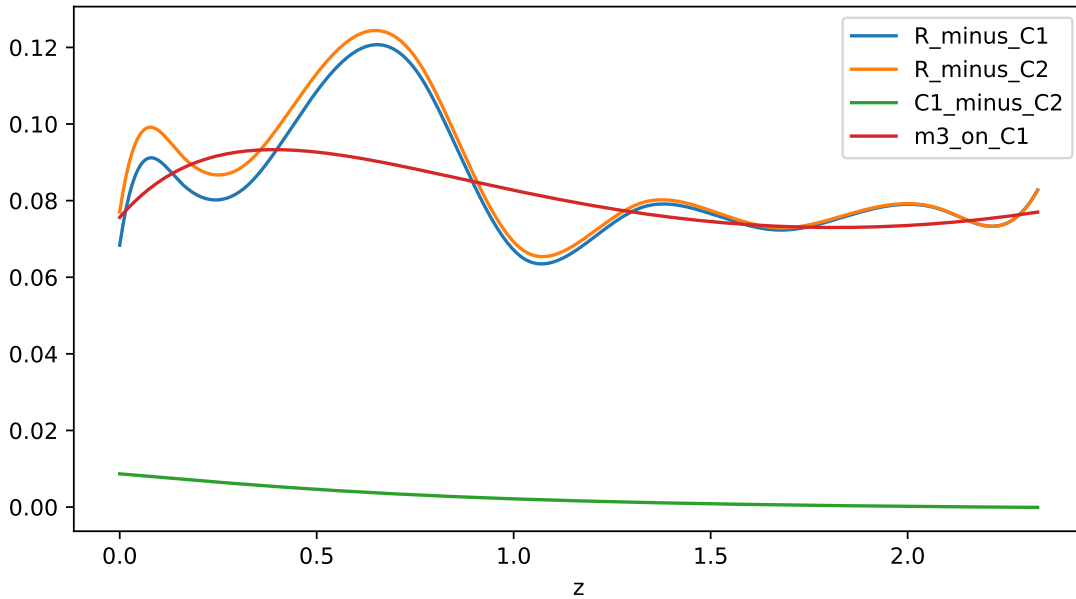


Figure 5: Reference-sensitivity overlay: the difference between R and C1 and the difference between R and C2 lie on top of one another across the constrained domain (the difference between R and C1 and the difference between R and C2 nearly coincide; the difference between C1 and C2 is much smaller), and the fixed order-3 reconstruction captures the broad low/intermediate-redshift trend but not the exact peak near $z \simeq 0.65$. Shading marks the constrained domain; the dotted line marks the Ly α extrapolation test.

6 Results

This section states the paper’s empirical results in one place. Construction details are in Sections 2–5. Headline numerical results are generated from or hash-bound to versioned publication evidence.

6.1 Headline endpoint separation

The phenomenological low-redshift distance-sector endpoint R ($H_0 = 72.80 \text{ km s}^{-1} \text{ Mpc}^{-1}$) differs substantially from both Planck-compatible references C1 ($H_0 = 67.99 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and C2 ($H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, secondary sensitivity reference) across the constrained domain $z \leq 1.8$. The difference between R and C1 and the difference between R and C2 Δ_1 and Δ_2 (defined in Section 2.4) share one dominant common mode (common-mode fraction ≥ 0.998 per interval), while the difference between C1 and C2 difference Δ_C is much smaller throughout: across the constrained bins, the RMS difference between C1 and C2 is approximately 1.3% to 8.4% of the corresponding RMS difference between R and either Planck reference .

6.2 Compact reconstruction

A five-coordinate order-3 map gives a compact empirical description of the constrained-domain ($z \leq 1.8$) endpoint difference with local projected Jacobian rank 5 of 5 fitted coordinates ($\chi^2 = 1459.96$, only 8.04 above the 21-coordinate control; AIC ≈ 1469.96 against ≈ 1493.92 for the control, reported descriptively as parsimony rather than evidence). Selection is predictive (held-out supernova and DESI $z \leq 1.8$ folds, Section 4) and carries no significance of its own: the matched constrained-domain null bootstrap tests representation complexity, not existence of the endpoint separation (null bootstrap $P(\text{select} \geq m=3) = 0.216$). Zero ruler-clock split is allowed ($\Delta\chi^2(\epsilon=0) = 0.124$). The fitted difference reaches its largest amplitude near $z \approx 0.65$ (Section 5.3).

6.3 Reference sensitivity

The fixed order-3 basis transfers across the two Planck references with small residual degradation (fraction 0.068): parameters learned on one reference construction predict the other separation. The difference between R and C1 and the difference between R and C2 share one dominant direction (cosine 0.9997, principal angle 1.5°). This reference-sensitivity check shows the deformation is insensitive to the tested CMB-reference construction; it does not assign a frequentist p-value to the similarity (Section 5.5). High-redshift continuation through the Lyman-alpha anchor remains unvalidated.

7 Discussion

7.1 What appears robust

Within the tested reconstructions over $z \leq 1.8$, six features recur. First, a common R-versus-CMB deformation: the difference between R and C1 and the difference between R and C2 lie on top of one another (common-mode fraction ≥ 0.998 per interval). Second, a peak near $z \approx 0.65$ (largest amplitude). Third, low/intermediate- z concentration: both the deformation RMS (peak bin $0.4 < z < 0.8$) and the fit-improvement attribution (peak bin $0 < z < 0.4$) concentrate below $z \sim 0.8$, though their peak bins differ. Fourth, one dominant shared direction (cosine 0.9997, principal angle 1.5° , SVD dominance 76.3). Fifth, a compact representation: five coordinates suffice ($\chi^2 = 1459.96$, local projected Jacobian rank 5 of 5 fitted coordinates, AIC ≈ 1469.96 against ≈ 1493.92 descriptively). Sixth, reference insensitivity across C1/C2: cross-reference transfer degrades by fraction 0.068 or less while the difference between C1 and C2 stays much smaller (approximately 1% to 8% of the corresponding R-versus-CMB difference across the constrained bins).

7.2 Possible physical interpretations

Candidate explanations fall into five categories: evolving late-time background dynamics reproducing the same $H(z)$ deformation; early-universe and ruler physics, including recombination or drag-epoch effects shifting the acoustic scale; combined early- and late-time effects; modified gravity and modified propagation, including distance-duality or luminosity-distance effects mimicking expansion structure; and conventional systematics or calibration effects in one or more of the distance, BAO, or CMB sectors.

7.3 What would discriminate interpretations

Untouched sectors would discriminate where background geometry cannot. Independent tests include cosmic chronometers, redshift-space distortions, weak and CMB lensing, time-delay distances, and standard sirens. No fitting in those sectors is performed in this paper.

8 Limitations

All reported values are best fits; no posterior intervals are claimed. Endpoint posterior uncertainty is not propagated into the difference fields. C1 and C2 are not statistically independent constructions: both use Planck information. R carries $r_d = 135.39$ Mpc while the C1/C2 references use recomputed CAMB rulers near 147.1 Mpc (Table 2). The constrained reconstruction domain is $z \leq 1.8$; the order-3 continuation fails at the $z = 2.33$ Ly α anchor. The calibrator result covers the declared contract only ($\Delta\mu_{\text{cal}} = 0$, a qualitative assumption, not quantitatively established).

Table 5: Limitations and current status.

Issue	Current status
Endpoint posterior uncertainty	Not propagated
C1/C2 statistical independence	Shared Planck information
Order-selection bootstrap	$P = 0.216$
Constrained reconstruction range	$z \leq 1.8$
Ly α continuation test	Fails at $z = 2.33$

The distance-sector endpoint and the Planck-compatible references prefer quantitatively different expansion histories (H_0 72.80 vs 67.99, with endpoint-specific acoustic rulers per Table 2).

8.1 Relation to neighbouring approaches

DESI DR2 BAO remain compatible with flat Λ CDM in isolation, while combinations with CMB and supernovae can favor evolving dark energy of the $w_0 w_a$ form [5, 15, 16]. Our reconstruction is not a dark-energy equation of state: it is a reconstruction of the required observational transformation between two calibrated histories, with dimensionality selected through held-out data rather than by allowing enough function freedom to improve the joint fit. That selection discipline is also what separates this work from unconstrained Gaussian-process or smoothing reconstructions of the expansion history [17] and from generic cosmography [18], which do not penalize flexibility by prediction. Early-dark-energy solutions modify the pre-recombination universe and its perturbations; the reconstruction assumes no recombination change and carries no perturbation sector. For the broader solution landscape we refer to the community review [19]. This work does not introduce another equation-of-state fit. It compares separately calibrated histories and asks whether

their difference is compact and reference-insensitive within the two tested Planck-compatible constructions. This decomposition is complementary to the scale-vs-shape obstruction argument [20]: uniform sound-horizon rescaling changes scale but cannot remove the BAO–SN shape disagreement, and our R endpoint contains both a large ruler shift ($\sim 8\%$) and a redshift-dependent shape difference.

9 Conclusion

Over $z \leq 1.8$, the distance-sector endpoint differs from both Planck references in nearly the same redshift-dependent way. Across the constrained bins, the RMS difference between C1 and C2 is approximately 1% to 8% of the corresponding difference between R and either Planck reference. The difference between R and C1 and the difference between R and C2 share one dominant direction (cosine 0.9997, principal angle 1.5°), with cross-reference transfer degradation fraction 0.068 or less, reaching its largest amplitude near $z \sim 0.65$.

A five-coordinate order-3 reconstruction summarizes this difference with local projected Jacobian rank 5 of 5 fitted coordinates ($\chi^2 = 1459.96$). The matched constrained-domain bootstrap gives $P(\text{select} \geq 3) = 0.216$, and the separate continuation to the $\text{Ly}\alpha$ anchor at $z = 2.33$ fails.

The resulting empirical target is the redshift-dependent difference between the distance-sector history and the Planck-calibrated histories over $z \leq 1.8$. Its compact form now provides a concrete expansion-history pattern against which candidate physical explanations and independent observables such as growth, lensing, chronometers, time-delay distances, and standard sirens can be tested.

A Likelihood details and calibration contract

The joint likelihood sums DESI DR2 BAO terms with published covariances, the compressed CMB prior used during endpoint discovery, and the Pantheon+SH0ES joint term. No external H_0 prior enters the likelihood.

Pantheon+SH0ES is one 1657-row covariance likelihood with one common profiled M . With $r_0 = m_B^{\text{corr}} - \mu_{\text{no } M}$,

$$\hat{M} = \frac{\mathbf{1}^T C^{-1} r_0}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad \chi_{\text{joint}}^2 = (r_0 - \hat{M} \mathbf{1})^T C^{-1} (r_0 - \hat{M} \mathbf{1}), \quad (13)$$

where C is the full 1657×1657 covariance. Ladder and non-ladder blocks are diagnostics: each uses its marginal covariance with an independent M , so $\chi_{\text{ladder}}^2 + \chi_{\text{non-ladder}}^2 \neq \chi_{\text{joint}}^2$.

Calibrator rows use $\mu_{\text{cal}} = \text{CEPH_DIST}$. Under the adopted calibration contract, no additional theory-side correction is applied to local calibrator distance moduli: $\Delta\mu_{\text{cal}} = 0$; this covers the declared contract only and is a qualitative assumption, not quantitatively established. Hubble-flow rows use z_{HD} for D_M and the $z_{\text{HD}}/z_{\text{HEL}}$ prefactor $D_L = (1 + z_{\text{HEL}})D_M(z_{\text{HD}})$, which assumes massless photons, null geodesics, and photon-number conservation [12].

The endpoint-discovery CMB block is the compressed set $(R, \ell_A, \omega_b h^2)$ with numerical values from [8], validated there against conventional late-time models only. Endpoint R was originally selected using the Chen et al. compressed Planck distance prior. Because those priors were validated for conventional late-time models, they are not used here as evidence that R's nonstandard ruler is physically viable. The C1/C2 reference constructions are calibrated directly to the full Planck spectra (Section 2.5).

A.1 Calibration contract

No telescope or catalog data were rescaled. All changes are in the forward model, not the data: observed redshifts, magnitudes, covariances, masks, and Cepheid distances are untouched.

DESI path: $z_{\text{obs}} \rightarrow z_{\text{bg}} \rightarrow D_M, D_H, D_V \rightarrow D/r_d$ with each endpoint's own ruler from Table 2; observed D/r_d rows and published covariances are unchanged. Pantheon+SH0ES path: 1657 selected rows including 77 calibrators; z_{HD} for cosmological distance, z_{HEL} in the luminosity prefactor, CEPH_DIST for calibrator μ ; one common profiled M with full 1657×1657 covariance in the fit; source-defined ladder split is diagnostic-only with independent M terms and is non-additive under the joint cross-covariance.

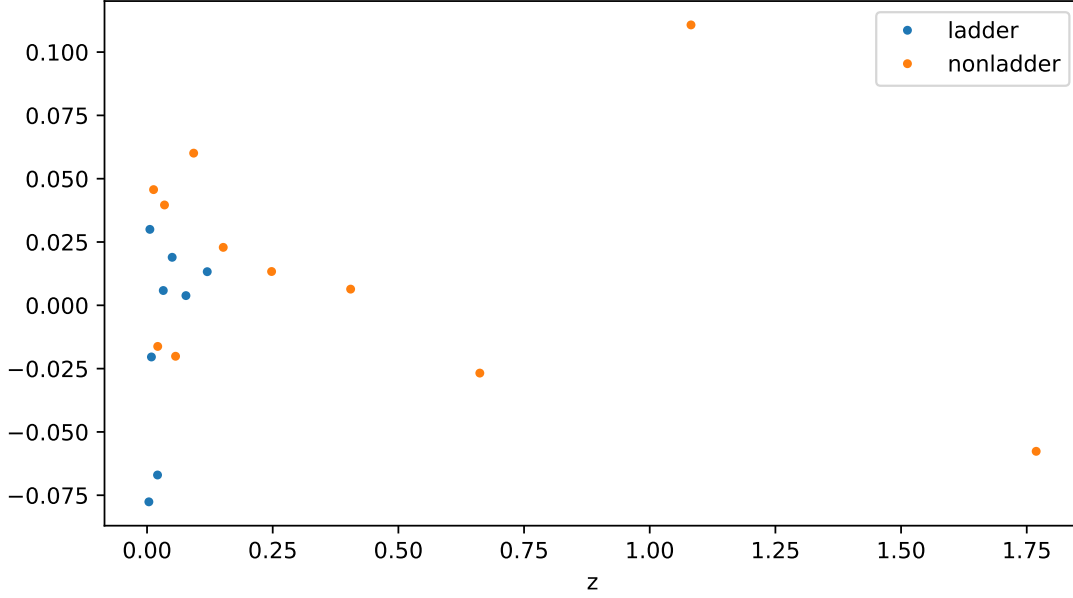


Figure 6: Appendix-only subset-profiled supernova diagnostics: source-defined ladder diagnostic, independently profiled M (left) and complementary non-ladder diagnostic, independently profiled M (right). Non-additive; never used in the joint fit.

B Reproducibility

The driver `scripts/paper/build_expansion_history_paper.py` replays the versioned publication evidence, rebuilds the results registry, regenerates all figures and tables with source-hash binding, validates claims in release mode, compiles the manuscript with `pdflatex` and `bibtex`, and builds and checks the arXiv bundle. The bundle `paper/dist/arxiv_submission/` is flat and self-contained, excluding run directories, source code, tests, credentials, logs, and caches. Random seeds, code hashes, and data hashes are recorded in the versioned publication evidence manifests.

Numerical stack. All array arithmetic uses NumPy [21] and all optimization and quadrature use SciPy [22]; versions are pinned in the versioned publication evidence manifests. Fixed endpoint artifacts and their hashes are listed in `paper/evidence_manifest.yaml`; regeneration fails on hash mismatch, so the published numbers reproduce exactly from the archived evidence. Every build step reads only files inside this public repository; no step reaches outside it.

B.1 Code and data availability

The paper source, derived publication evidence, reproducibility scripts, and interactive visualization are available at https://github.com/jenholm/physics_models/tree/main/redshift-dependent-expansion. Public evidence records include the R endpoint history (https://github.com/jenholm/physics_models/blob/main/redshift-dependent-expansion/paper/evidence/endpoints/endpoint_redshift.json), the C1 reference (https://github.com/jenholm/physics_models/blob/main/redshift-dependent-expansion/paper/evidence/endpoints/endpoint_cmb_1.json), the C2 reference (https://github.com/jenholm/physics_models/blob/main/redshift-dependent-expansion/paper/evidence/endpoints/endpoint_cmb_2.json), the constrained-domain definition (https://github.com/jenholm/physics_models/blob/main/redshift-dependent-expansion/paper/

evidence/endpoints/domain_split.json), the reference-sensitivity metrics (https://github.com/jenholm/physics_models/blob/main/redshift-dependent-expansion/paper/evidence/reference_sensitivity/scale_comparison.json and https://github.com/jenholm/physics_models/blob/main/redshift-dependent-expansion/paper/evidence/reference_sensitivity/agreement_metrics.json), and the model-selection and bootstrap evidence (https://github.com/jenholm/physics_models/blob/main/redshift-dependent-expansion/paper/evidence/model_selection/constrained_cv.json and https://github.com/jenholm/physics_models/blob/main/redshift-dependent-expansion/paper/evidence/model_selection/bootstrap.json).

References

- [1] Adam G. Riess, Wenlong Yuan, Lucas M. Macri, Dan Scolnic, Dillon Brout, Stefano Casertano, David O. Jones, Yukei Murakami, Gagandeep S. Anand, Caroline Huang, et al. A comprehensive measurement of the local value of the Hubble constant with $1 \text{ km s}^{-1} \text{ mpc}^{-1}$ uncertainty from the Hubble space telescope and the SH0ES team. *The Astrophysical Journal Letters*, 934: L7, 2022. doi: 10.3847/2041-8213/ac5c5b.
- [2] N. Aghanim et al. Planck 2018 results. VI. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020. doi: 10.1051/0004-6361/201833910.
- [3] Licia Verde, Tommaso Treu, and Adam G. Riess. Tensions between the early and the late universe. *Nature Astronomy*, 3:891, 2019. doi: 10.1038/s41550-019-0902-0.
- [4] Lloyd Knox and Marius Millea. Hubble constant hunter’s guide. *Physical Review D*, 101: 043533, 2020. doi: 10.1103/PhysRevD.101.043533.
- [5] DESI Collaboration. DESI DR2 results II: Measurements of baryon acoustic oscillations and cosmological constraints. *Physical Review D*, 112:083515, 2025. doi: 10.1103/tr6y-kpc6.
- [6] Dan Scolnic, Dillon Brout, Anthony Carr, Armin Rest, Adam G. Riess, Tamara M. Davis, Samuel R. Hinton, David O. Jones, Erik R. Peterson, Brodie Popovic, et al. The Pantheon+ analysis: The full data set and light-curve release. *The Astrophysical Journal*, 938:113, 2022. doi: 10.3847/1538-4357/ac8b7a.
- [7] Dillon Brout, Dan Scolnic, Brodie Popovic, Adam G. Riess, Anthony Carr, Joe Zuntz, Rick Kessler, Tamara M. Davis, Samuel R. Hinton, David O. Jones, et al. The Pantheon+ analysis: Cosmological constraints. *The Astrophysical Journal*, 938:110, 2022. doi: 10.3847/1538-4357/ac8e04.
- [8] Lu Chen, Qing-Guo Huang, and Ke Wang. Distance priors from Planck final release. *Journal of Cosmology and Astroparticle Physics*, 2019(02):028, 2019. doi: 10.1088/1475-7516/2019/02/028.
- [9] Antony Lewis, Anthony Challinor, and Anthony Lasenby. Efficient computation of cosmic microwave background anisotropies in closed friedmann-robertson-walker models. *The Astrophysical Journal*, 538:473–476, 2000. doi: 10.1086/309179.
- [10] DESI Collaboration. DESI DR2 results I: Baryon acoustic oscillations from the lyman alpha forest. *Physical Review D*, 112:083514, 2025. doi: 10.1103/2wwn-xjm5.
- [11] Adam G. Riess, Stefano Casertano, Wenlong Yuan, Lucas M. Macri, and Dan Scolnic. Large Magellanic cloud cepheid standards provide a 1% foundation for the determination of the Hubble constant and stronger evidence for physics beyond Λ CDM. *The Astrophysical Journal*, 876:85, 2019. doi: 10.3847/1538-4357/ab1422.
- [12] I. M. H. Etherington. On the definition of distance in general relativity. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 15:761–773, 1933. doi: 10.1080/14786443309462220.
- [13] Daniel J. Eisenstein and Wayne Hu. Baryonic features in the matter transfer function. *The Astrophysical Journal*, 496:605, 1998. doi: 10.1086/305424.

- [14] Wayne Hu and Naoshi Sugiyama. Small-scale cosmological perturbations: An analytic approach. *The Astrophysical Journal*, 471:542, 1996. doi: 10.1086/178032.
- [15] Michel Chevallier and David Polarski. Accelerating universes with scaling dark matter. *International Journal of Modern Physics D*, 10:213–224, 2001.
- [16] Eric V. Linder. Exploring the expansion history of the universe. *Physical Review Letters*, 90: 091301, 2003.
- [17] Arman Shafieloo, Ujjaini Alam, Varun Sahni, and Alexei A. Starobinsky. Smoothing supernova data to reconstruct the expansion history of the universe and its age. *Monthly Notices of the Royal Astronomical Society*, 366:1081–1095, 2006.
- [18] Celine Cattoen and Matt Visser. Cosmography: Comments on the “Jerk” paper. *Classical and Quantum Gravity*, 24:5985–5997, 2007.
- [19] Eleonora Di Valentino et al. In the realm of the Hubble tension — a review of solutions. *Classical and Quantum Gravity*, 38:153001, 2021.
- [20] Zhihuan Zhou, ShengYue Wang, Zhuang Miao, Chaoqian Ai, and Hongchao Zhang. Geometric obstruction to resolving the Hubble tension: Orthogonality of scale and shape in distance measurements. *Physical Review D*, 114:063511, 2026. doi: 10.1103/hv6b-kccr.
- [21] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.
- [22] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17: 261–272, 2020. doi: 10.1038/s41592-019-0686-2.